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BCA (Sem-IV) – Software Engineering (402) Core-IX

2024

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer all sections as directed.

Section–A

Objective type Questions

Compulsory

1. Choose the correct answer of the following multiple choice questions : 2×10=20

   (i) Which of the following best defines **Software Engineering as a Layered Technology?**

a) It is a collection of programming techniques

b) It consists of process, methods, and tools layered over a quality focus

c) It is an alternative to traditional engineering

d) It eliminates the need for process models

(ii) Which of the following software process models is best suited for projects with highly uncertain requirements?

a) Waterfall Model

b) Incremental Model

c) Spiral Model

d) Prototyping Model

(iii) The Capability Maturity Model Integration (CMMI) primarily focuses on:

a) Ensuring that software is defect-free

b) Establishing a structured approach to software process improvement

c) Reducing software development cost

d) Replacing traditional project management

(iv) In the software process framework, what is the main difference between framework activities and umbrella activities?

a) Framework activities are optional, while umbrella activities are mandatory

b) Framework activities define the core process, while umbrella activities provide support functions

c) Framework activities focus on testing, while umbrella activities focus on coding

d) There is no difference; both serve the same purpose

(v) Which of the following is NOT a requirement analysis technique?

a) Prototyping

b) Data Flow Diagrams (DFDs)

c) UML Sequence Diagrams

d) Code Review

(vi) Functional requirements in software engineering define:

a) The software's internal architecture

b) The system's expected behavior and features

c) The software's security policies

d) The constraints on design choices

(vii) Why is Flow-Oriented Modeling important in requirement analysis?

a) It helps in understanding the movement of data within a system

b) It provides a graphical user interface for the system

c) It eliminates the need for coding

d) It ensures faster development without documentation

(viii) Which of the following is an example of a risk mitigation strategy?

a) Identifying new risks after development

b) Ignoring minor risks to focus on major ones

c) Allocating additional resources to critical project tasks

d) Delaying risk management activities until the testing phase

(ix) What is the main purpose of a Risk Mitigation, Monitoring, and Management (RMMM) Plan?

a) To eliminate all project risks

b) To document how risks will be tracked and handled during development

c) To reduce project cost

d) To focus only on high-priority risks

(x) Which of the following best defines "Software Quality Assurance (SQA)"?

a) A method to test software before deployment

b) A process to ensure that software meets predefined quality standards

c) A documentation technique for software development

d) A way to reduce the cost of software production

Section–B

Short Answer type Questions

Answer any **four** questions of the following : 5×4=20

2. Define SDLC waterfall model.

3. Explain why software does not "wear out" like physical hardware but still requires maintenance.

4. Why is software engineering considered a layered technology?
5. Describe the major differences between functional and non-functional requirements with examples.
6. How does Risk Projection differ from Risk Identification in software engineering?
7. What are Data Flow Diagrams (DFD)? Draw 0 Level and 1 level DFD for Student Registration System.
8. Explain Peer Reviews in Software Engineering.
9. Why do software projects often exceed their estimated schedule despite proper planning?

Section–C

Long Answer type Questions

Answer any **two** questions of the following : 15×2=30

10. **Explain all five maturity levels in CMMI.**

11. Explain how PERT (Program Evaluation and Review Technique) is used for project scheduling.
12. What is the significance of a Risk Mitigation, Monitoring, and Management (RMMM) Plan?
13. Explain why ambiguity in Software Requirements Specification (SRS) can lead to project failure.

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